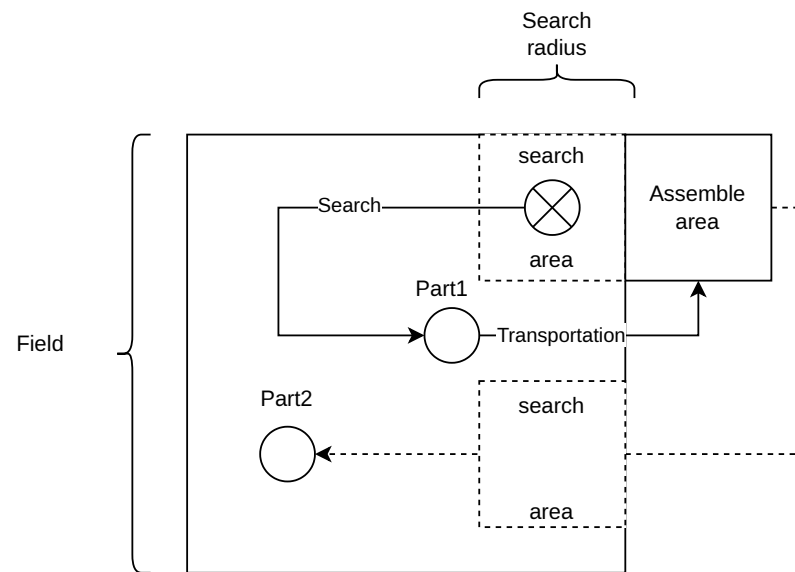
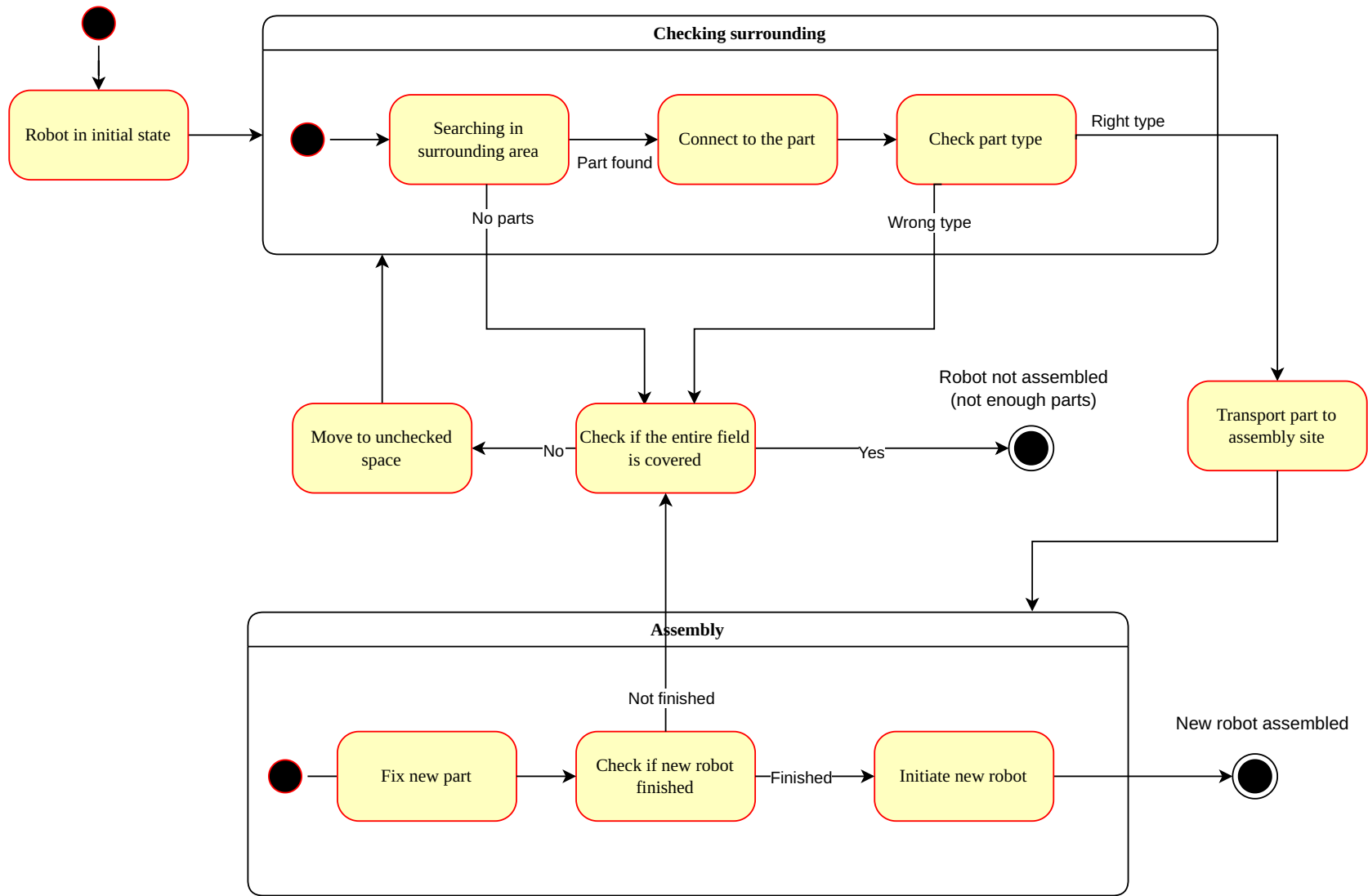


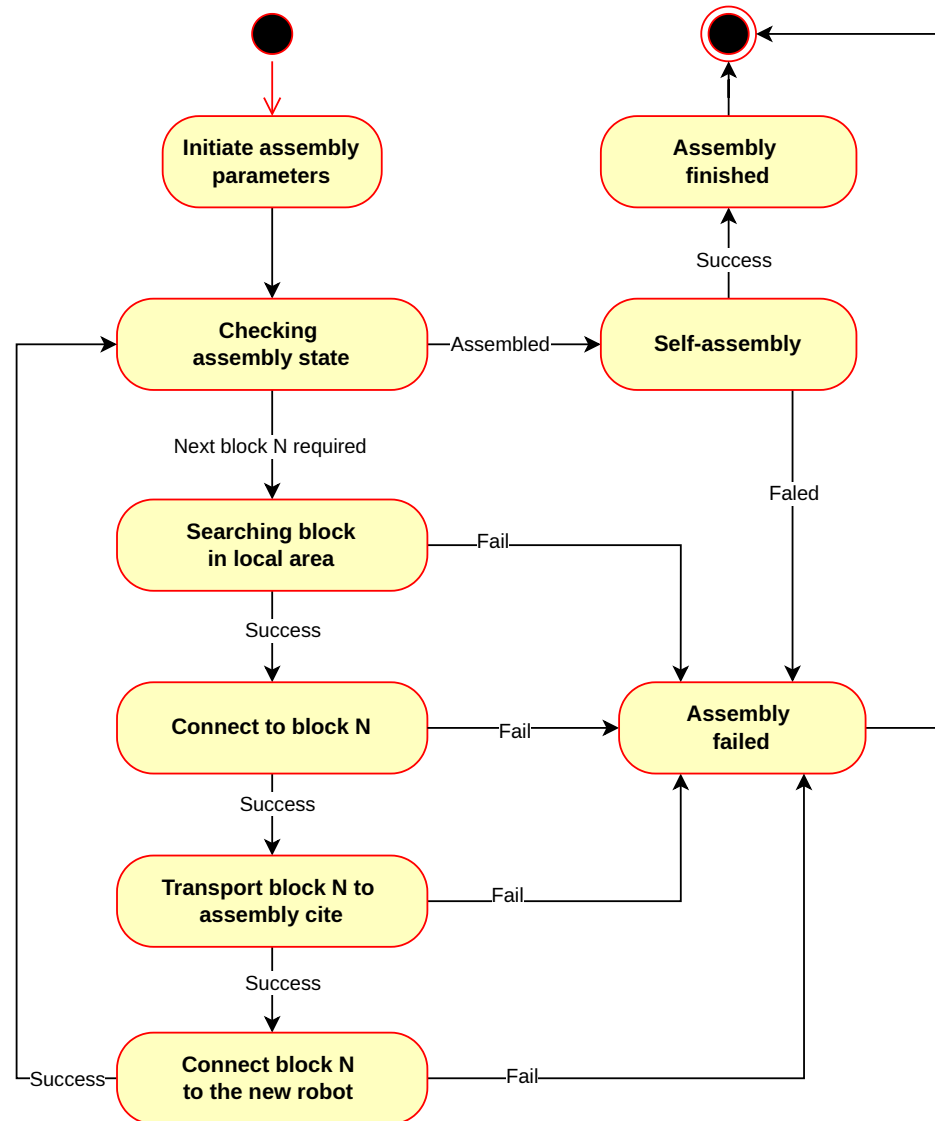
Robotic assemble system visualization



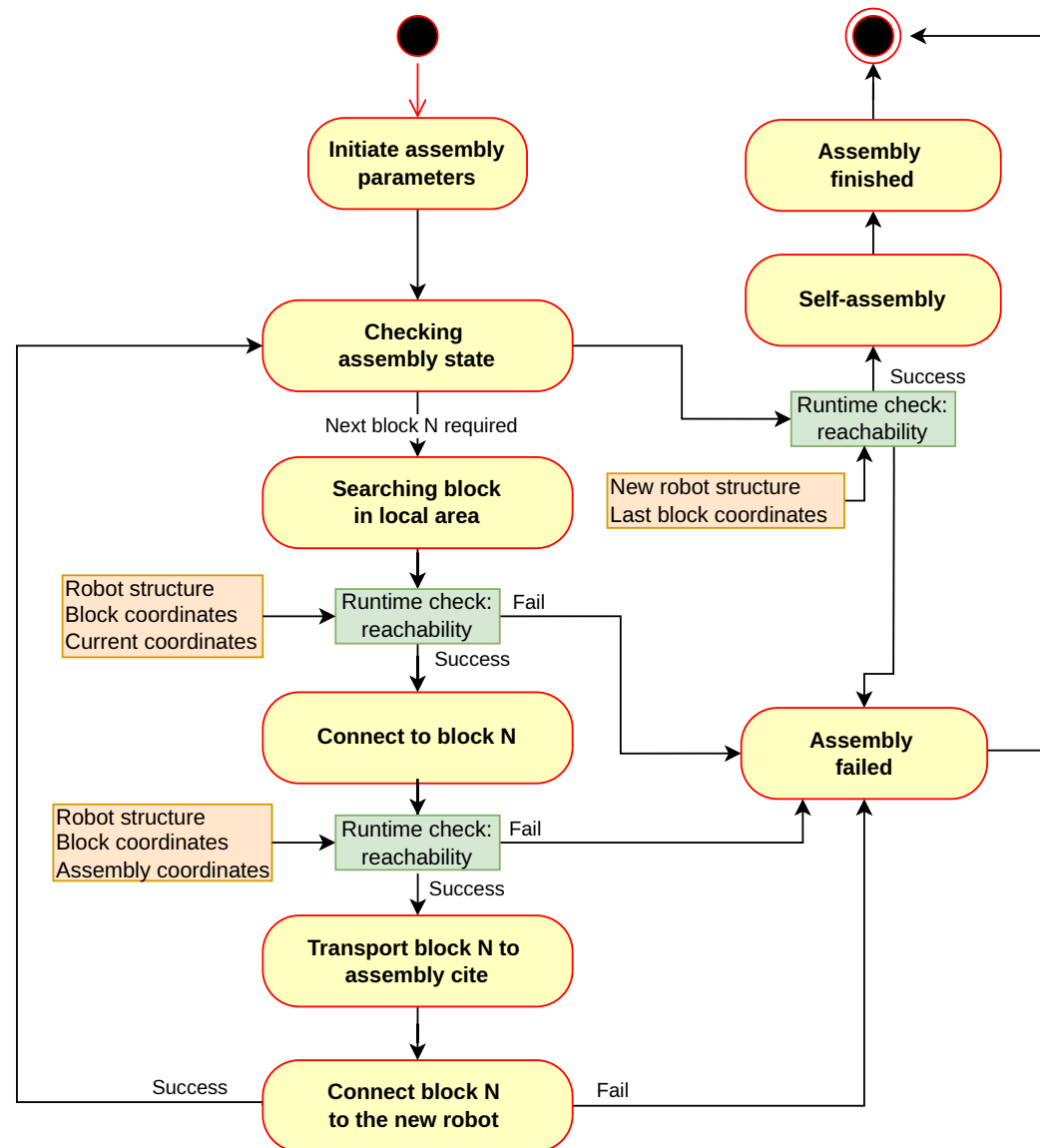
Robotic assemble system statechart



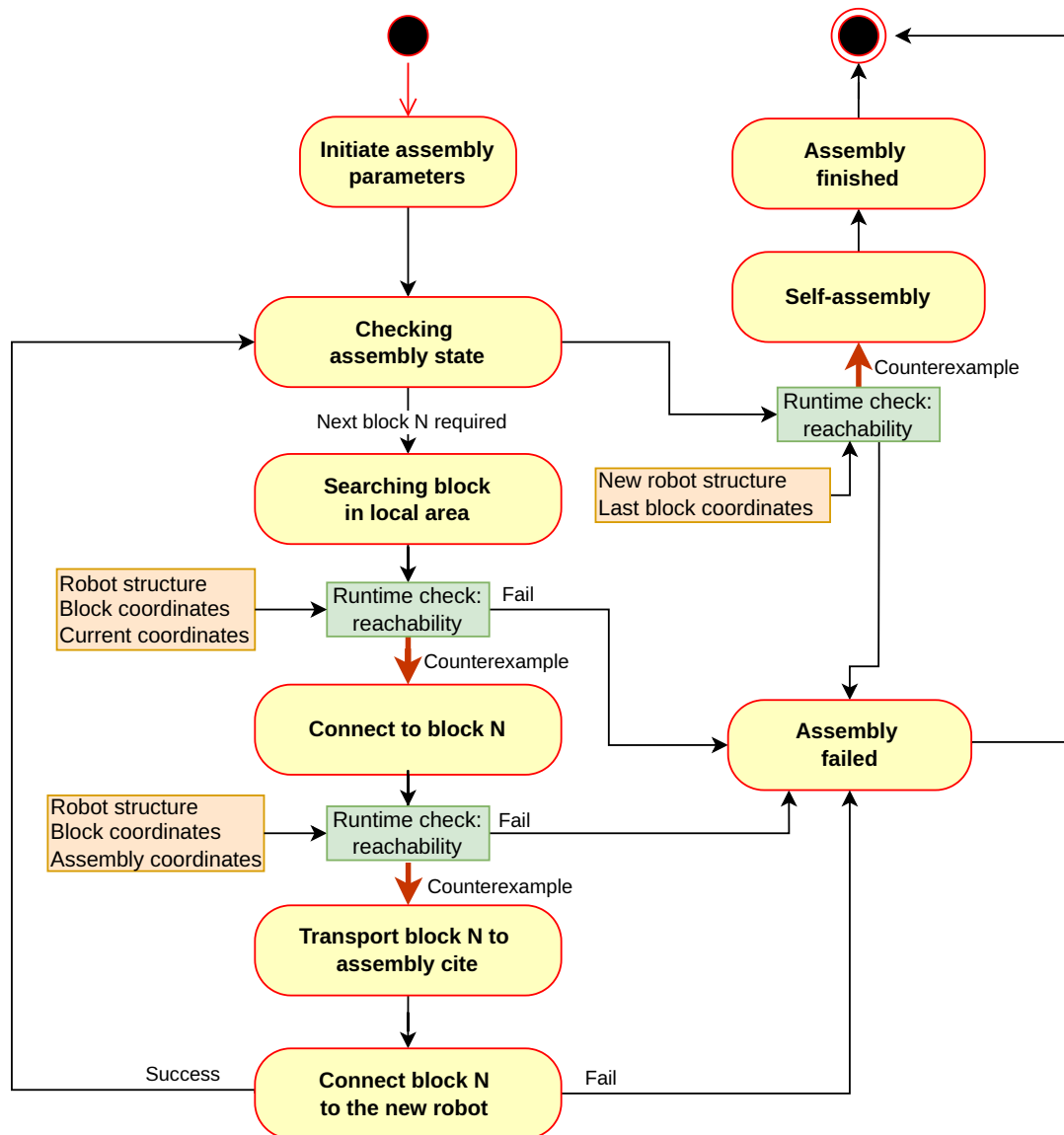
Local assembly



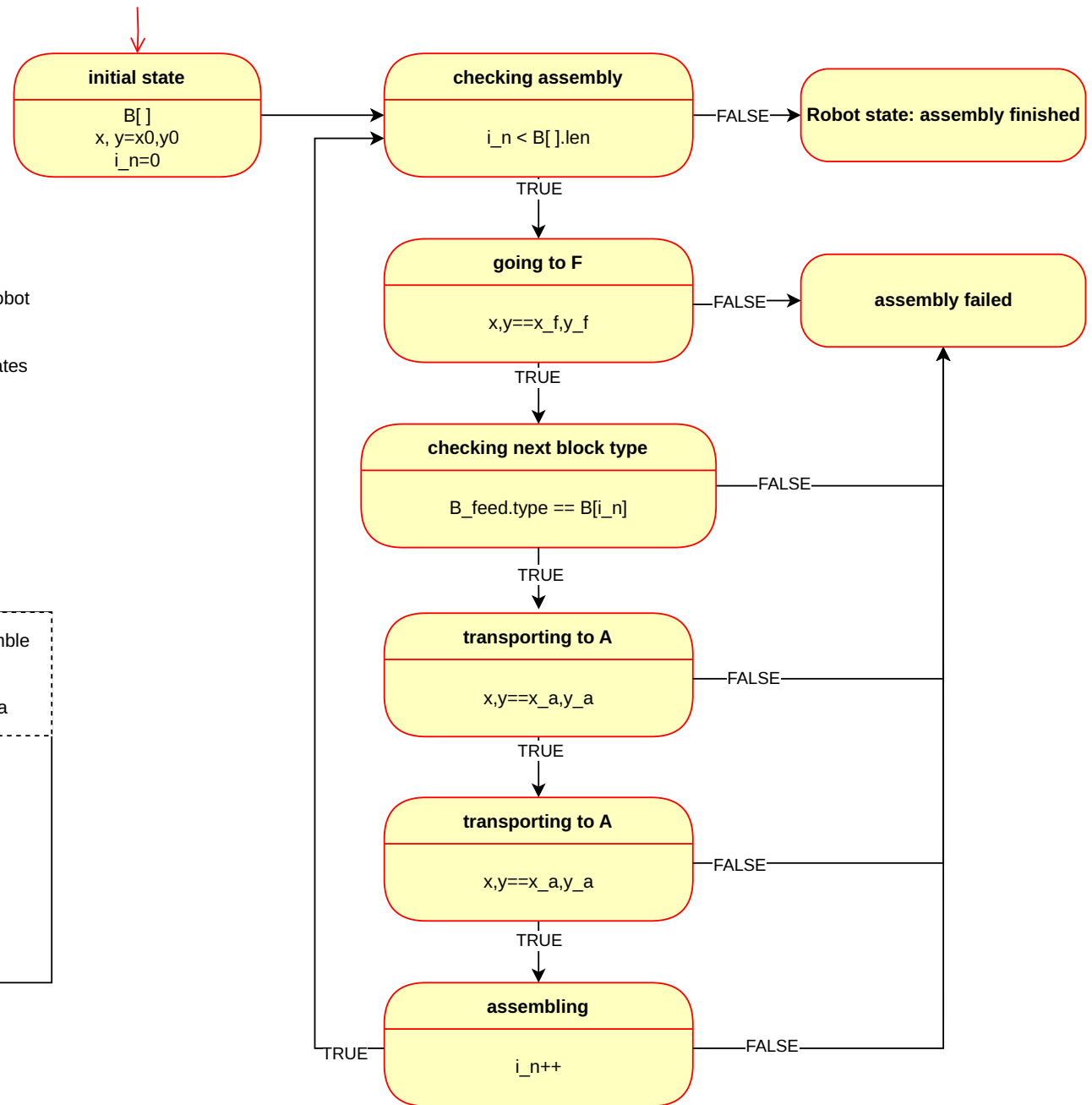
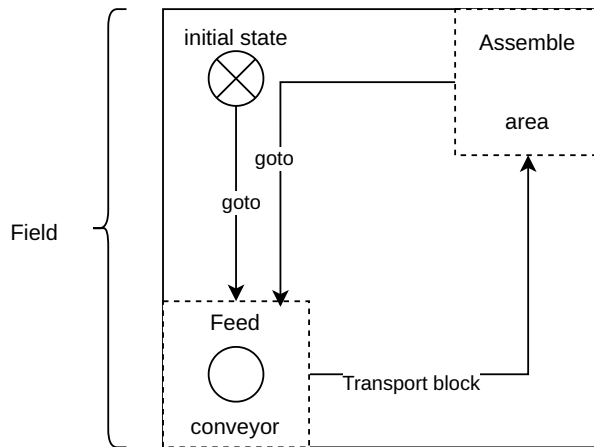
Local assembly with runtime model checking

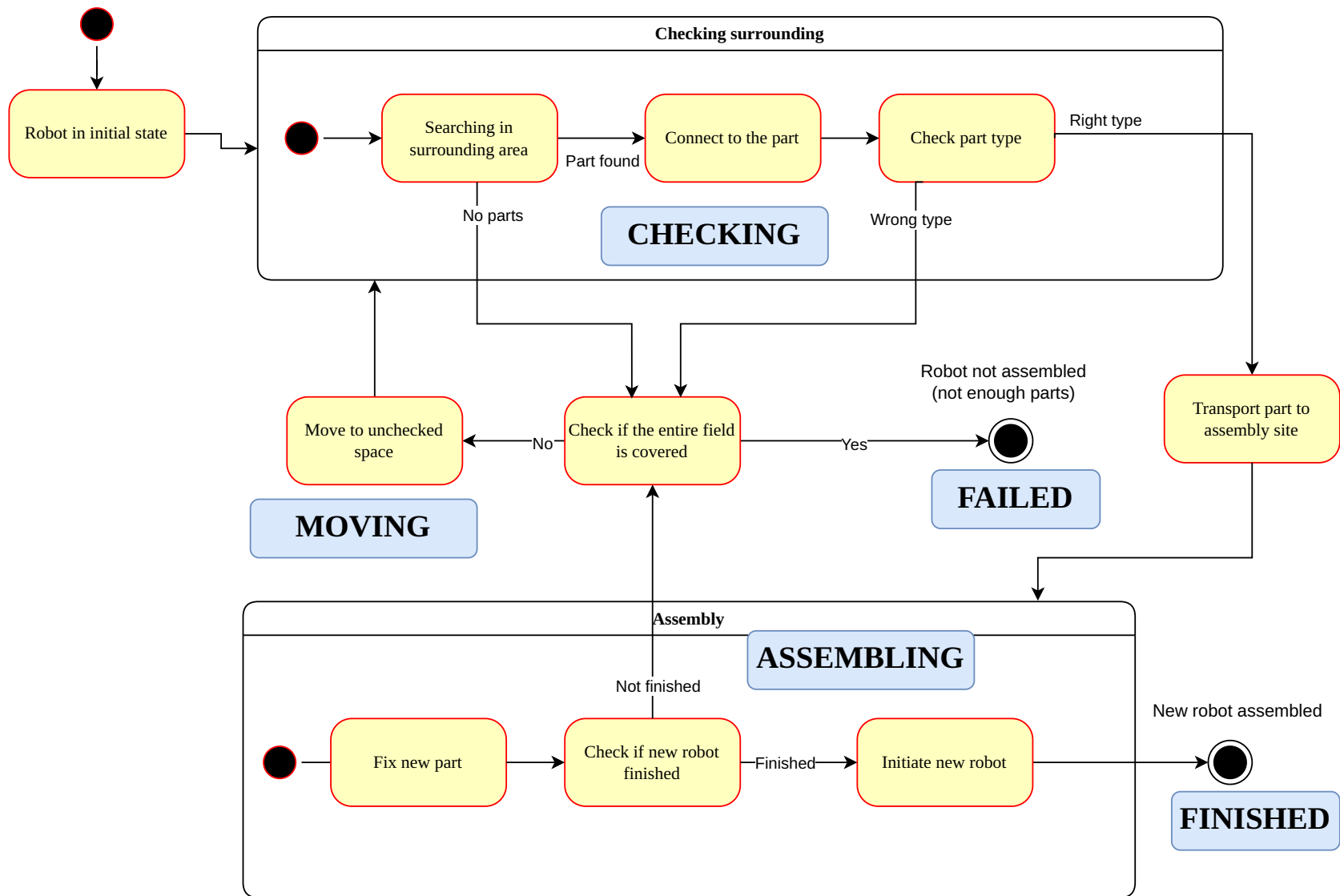


Local assembly with runtime model checking and counterexample based control



B[] - structure of new robot
 i_n - index of next block for new robot
 x,y - current robot coordinates
 x0,y0 - initial coordinates
 x_a,y_a - assembly point coordinates
 x_f, y_f - feed point coordinates
 A - assemble point
 F - feed point





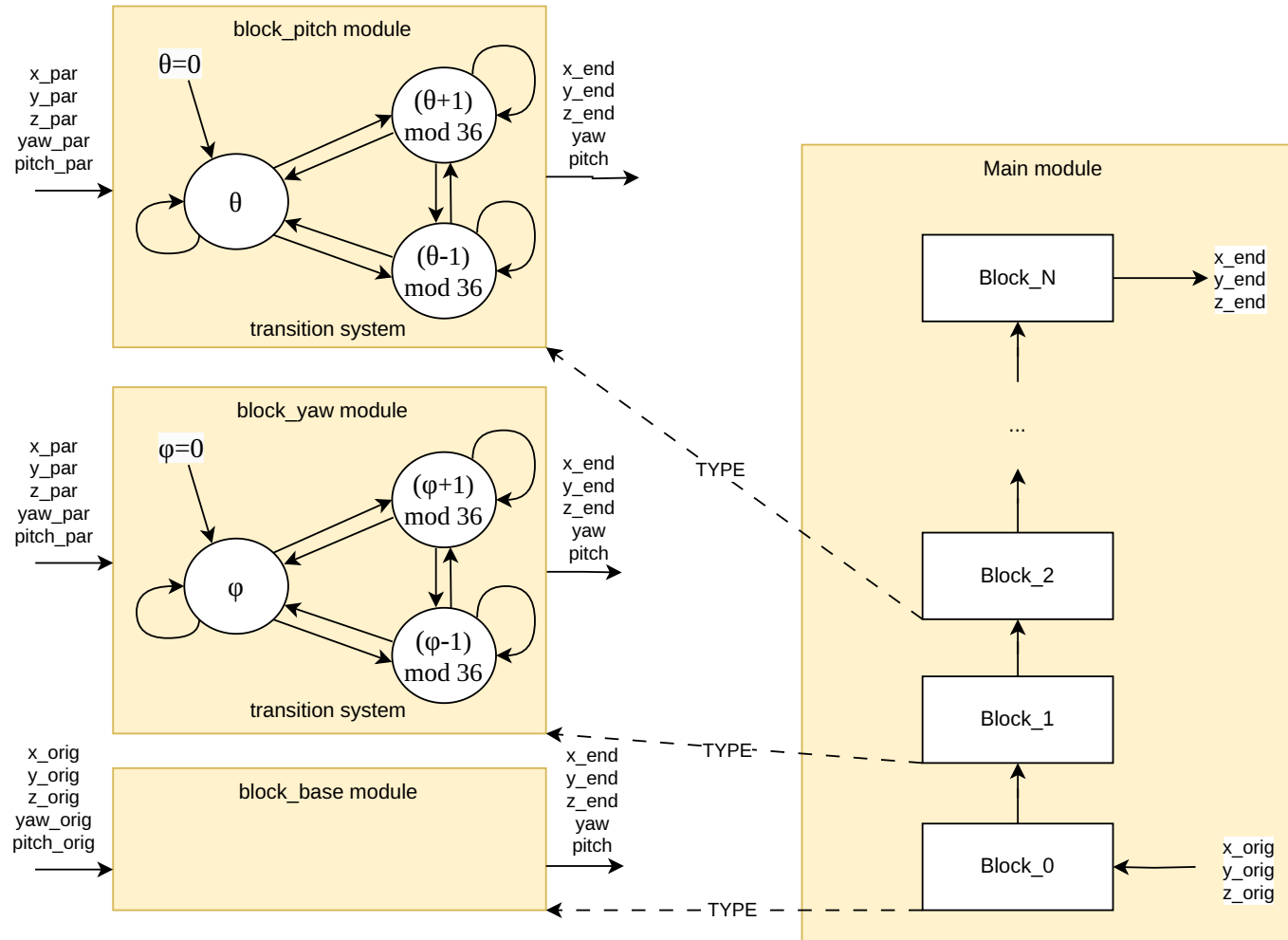
Local assembly

**Initiate assembly
parameters**

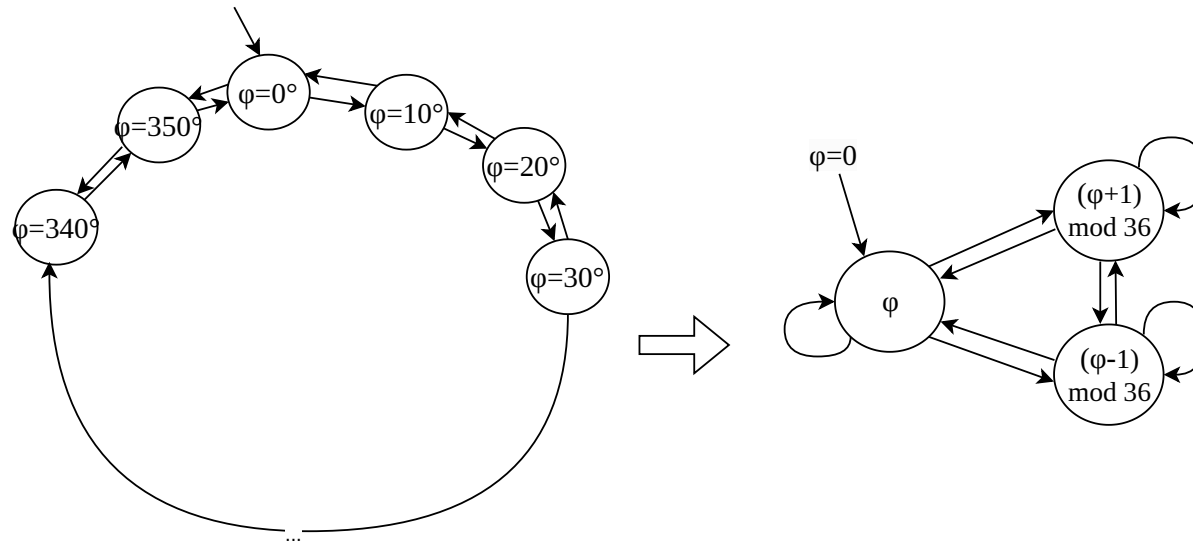
New_robot={Block1, Block2, ...}
Next_block = New_robot[0]
X_end = ..., Y_end = ...
X_asmb1 = ..., Y_asmb = ...



Robot configuration model checking

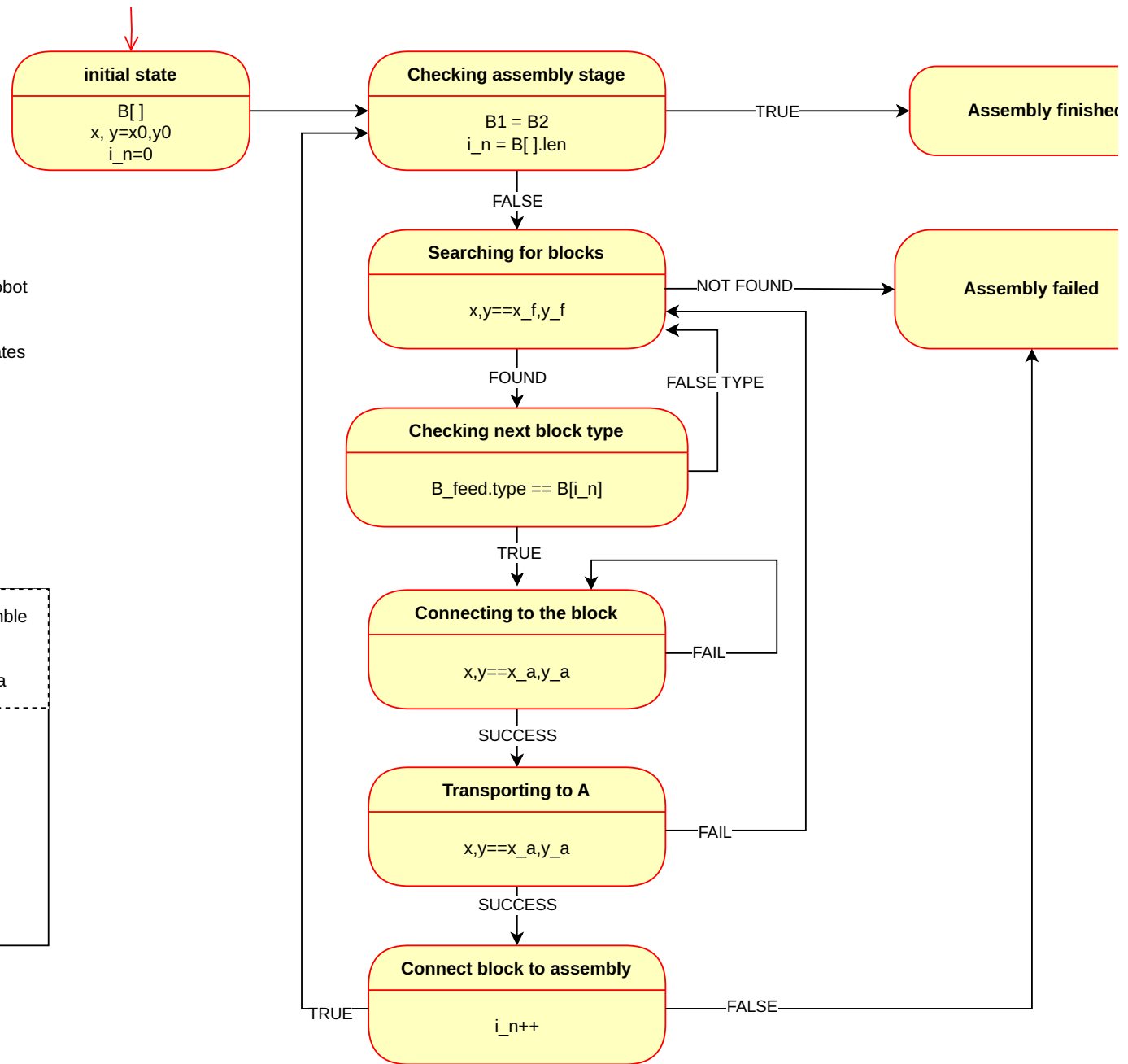
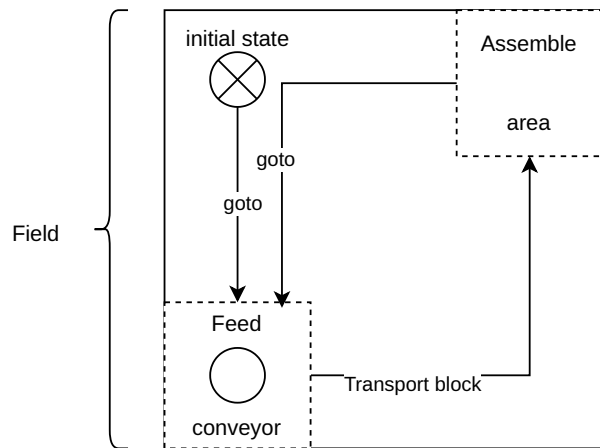


Robot block transition system

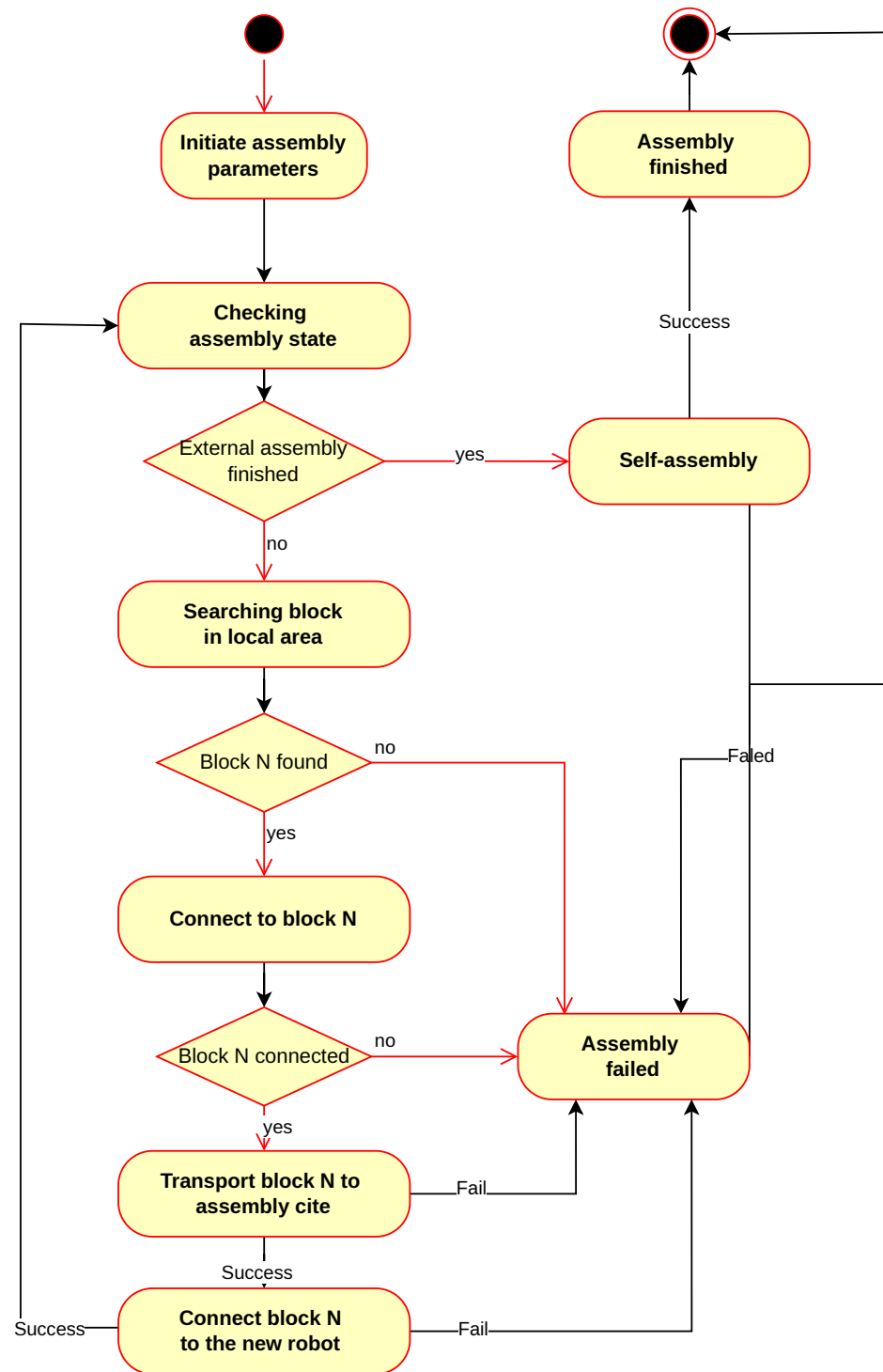


Local assembly

B[] - structure of new robot
 i_n - index of next block for new robot
 x,y - current robot coordinates
 x0,y0 - initial coordinates
 x_a,y_a - assembly point coordinates
 x_f,y_f - feed point coordinates
 A - assemble point
 F - feed point



d



Robot configurations model checking

